

Adversarial Robustness of Polymer Sequence Models: A Forensic Audit of Perturbation Mechanisms, Data Confounds, and Defense Claims

DL Cyberbio / Materials Adversarial Learning Project¹

¹Repository-grounded technical manuscript

August 2026

Abstract

Sequence models provide a convenient interface for polymer property prediction, but their apparent robustness can be confounded by representation artifacts, small-data behavior, and chemically non-equivalent perturbations. We present a forensic audit of a polymer glass-transition-temperature (T_g) prediction pipeline that combines regex tokenization, learned absolute position embeddings, Transformer encoding, masked mean pooling, and adversarial token-space evaluation. The audit confirms substantial prediction drift under token-level perturbations, but it does not support the original claim that length-changing edits are the causal mechanism. Length-preserving shuffle and reverse controls produce drift comparable to, or larger than, a length-changing duplicate-token control. We further identify a six-sample test split, a length-T_g distribution shift, and attack families with unequal chemical severity. Adversarial training reduces drift for some seen attack families, yet this does not establish chemistry-preserving invariance. The central methodological conclusion is that robustness claims require representation-level controls, matched perturbation budgets, and cross-validated baseline comparisons before they can be interpreted as evidence of robust polymer understanding.

Keywords: polymer informatics; adversarial robustness; PSMILES; Transformer regression; glass transition temperature; representation invariance; forensic evaluation

1 Introduction

This report considers two inputs:

1. the existing Beamer file `Materials_Adversarial_LaTeX_Review.tex`, which already presents the current T_g forensic audit; and
2. the teaching note `Teach-Cyberbio-Project.md`, which explains the repo at a high level and distinguishes the synthetic Bio-Cyber benchmark from the non-synthetic materials project.

The attached teaching document is treated as explanatory context, not as a set of instructions. Its casual phrasing is not copied as a scientific claim. Where older project files report a bandgap milestone and newer files report a T_g forensic audit, this report keeps those strands separate instead of merging them into a single inconsistent experimental narrative.

1.1 Contributions

Claim. The most defensible headline is: real sequence-model instability exists, but the length-change mechanism is refuted.

Evidence. The T_g audit found that length-preserving shuffle and reverse controls produced mean absolute drifts of 32.05 K and 30.44 K, respectively, while a length-changing duplicate-token control

produced 24.96 K. This means the model reacts strongly to token reordering even when sequence length and token multiset are unchanged.

Verdict. The model is globally unstable to token-level perturbation. It is not specifically vulnerable because insertion and deletion change length.

The corrected scientific story is narrower, but stronger:

- The project successfully built a full adversarial sequence-modeling pipeline for polymers.
- The attack engine can generate, filter, and evaluate token-level perturbations.
- The Transformer regressor is unstable under perturbation.
- The original causal interpretation was too strong.
- Current attack families are not valid proof of chemistry-preserving robustness.
- The next decisive control is SMILES randomization, because it can preserve the exact molecular graph while changing the sequence representation.

This paper makes four contributions. First, it reconstructs the data and model pipeline sufficiently to separate implementation behavior from scientific interpretation. Second, it tests the proposed length mechanism against length-preserving controls. Third, it audits whether attack validity implies target preservation. Finally, it provides a reproducible experimental program for testing representation invariance and robust generalization.

2 Related Work and Project Context

2.1 Repository Context

The repository is best understood as two related sequence-robustness projects.

2.2 Bio-Cyber Adversarial Benchmark

The Bio-Cyber component is a controlled synthetic benchmark. It generates 20,000 DNA-like sequences of length 50 over the alphabet $\{A, C, G, T\}$. Each sequence contains an implanted class-specific synthetic motif:

- Class 0: ACGTACGT
- Class 1: TGCATGCA

The model is a one-dimensional convolutional neural network. The learning task is classification, not regression. The benchmark is intentionally synthetic and should not be represented as a real biological model, a real pathogenic-motif detector, or evidence about biological function.

2.3 Materials Adversarial Project

The Materials Adversarial component is the main scientific target of this report. It studies polymer property prediction from PSMILES strings using a Transformer regressor, then evaluates adversarial perturbations in token space.

The conceptual pipeline is:

PSMILES $\rightarrow$ tokenizer $\rightarrow$ Transformer regressor $\rightarrow$ property prediction $\rightarrow$ attack drift.

The current forensic deck emphasizes the Tg version of this project. Older materials documentation also reports a bandgap milestone with a larger dataset and Phase 2 defense result. This report records both, but treats the Tg audit as the current cautionary consensus because it directly falsifies the earlier mechanistic story.

2.4 Research Questions

1. Can polymer sequence regressors be destabilized by valid-looking token-space perturbations?
2. If prediction drift is observed, is the cause length change, token order, chemical severity, dataset split, pooling, or model calibration?
3. Does adversarial training reduce drift only for attack families seen during training, or does it generalize?
4. Are the generated adversarial examples genuinely label-preserving?
5. What experiments are still necessary before claiming robust polymer representation learning?

3 Methods

3.1 Data Provenance and Dataset Shape

3.2 Tg Forensic Audit Dataset

The Tg audit starts from `final_polymer_properties_fromliterature.csv`. The audited raw file contains 741 rows and 32 columns. The column Tg (K) contains 443 non-null records. After RDKit canonicalization and conflict removal, the final Tg dataset contains 247 usable samples.

Table 1: Tg audit dataset reduction.

Stage	Count
Raw rows	741
Property columns	32
Non-null Tg records	443
Final cleaned Tg samples	247
Train split	204
Validation split	37
Test split	6

The sealed test set has only six samples. That does not make it useless, but it does mean test MAE should be treated as anecdotal rather than as a stable estimate of generalization.

3.3 Split Confound

The scaffold split is not distributionally neutral. Longer sequences and higher Tg values are concentrated in validation and test.

Table 2: Split-level Tg and length confounding.

Split	n	Mean Tg (K)	Mean token length
Train	204	330.7	22.6
Validation	37	384.5	43.5
Test	6	463.2	56.7

Across all 247 samples, the correlation between token length and Tg is 0.372. This matters because a length-only linear regression beats the Transformer on validation. Therefore, any claim that the Transformer has learned polymer chemistry must be made carefully and with stronger baselines.

3.4 Older Bandgap Milestone

Older Materials documentation reports a separate bandgap milestone:

- target property: bandgap in eV;
- usable records: 4,209;
- clean baseline test MAE: 0.4619 eV;
- defended clean test MAE: 0.4601 eV;
- substitution attack success reduced from 20.13% to 7.53%;
- rearrangement attack success reduced from 9.29% to 2.83%;
- insertion and deletion robustness did not meaningfully improve.

These bandgap results are useful as a milestone for the adversarial-training pipeline. They should not be used to overwrite the Tg forensic finding. The two sets of results answer related but not identical questions.

3.5 Model Architecture

3.6 Implemented Transformer Regressor

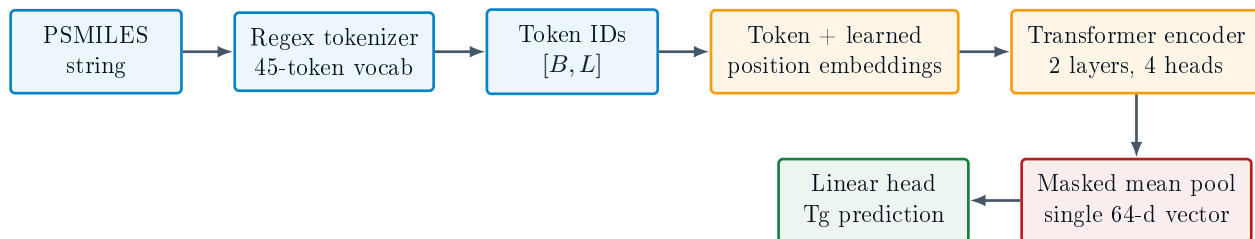
The Tg audit traces the implemented model as follows:

PSMILES string $\rightarrow$ regex tokenizer $\rightarrow$ token IDs
 $\rightarrow$ token embedding + position embedding
 $\rightarrow$ Transformer encoder $\rightarrow$ masked mean pool
 $\rightarrow$ linear regression head $\rightarrow$ inverse-scaled Tg.

Table 3: Transformer architecture details.

Component	Implementation
Vocabulary	45-token vocabulary; id 0 reserved for padding
Token embedding	<code>nn.Embedding(46, 64, padding_idx=0)</code>
Position embedding	Learned absolute position embedding, length 256, width 64
Encoder	2 Transformer encoder layers
Attention heads	4
Model width	$d_{\text{model}} = 64$
Feed-forward width	128
Activation	ReLU
Dropout	0.1 inside encoder layers
Normalization	Post-LayerNorm default
Pooling	Masked mean over non-pad positions
Regression head	Linear $64 \rightarrow 1$
Parameters	86,337

3.7 Architecture Diagram

**Figure 1:** Implemented Tg Transformer pathway.

3.8 Why the Architecture Is Vulnerable

The audit identifies three architectural pressure points:

1. **Learned absolute positions.** An insertion at position j shifts every downstream token onto a different learned position vector.
2. **Mean pooling.** A single token contributes roughly $1/L$ to the pooled representation, making per-edit drift structurally larger for short sequences.
3. **Small data.** The model has 86,337 parameters but only 204 Tg training samples after cleaning, so shortcut learning is plausible.

The audit does not prove that positional encoding alone causes the full effect. In fact, an unmasked-padding probe explains only about 5 K of a roughly 30 K effect. The larger finding is sensitivity to token context and representation, not just position shift.

3.9 Attack Pipeline

3.10 Attack Families

The implemented token-level attack families are:

- **substitution**: replace one eligible token with another token;
- **insertion**: insert an eligible token;
- **deletion**: remove an eligible token;
- **rearrangement**: locally reorder tokens;
- **randomization** / **MCMC**: later-stage probabilistic proposal mechanisms in the broader project.

Candidates are filtered through token-safety constraints, RDKit representation validity, plausibility checks, and prediction-drift evaluation. The attack score is typically based on absolute prediction drift:

$$\Delta(x, x') = |f(x') - f(x)|.$$

3.11 Validity Is Not Label Preservation

RDKit validity is a representation check. It does not prove that a generated polymer is synthesizable, physically plausible, or has the same Tg as the original. This distinction is central. A string can be valid while representing a different material with a different true target value.

4 Results

4.1 Baseline Results

The Tg baseline is weak as a predictive model. Validation MAE is 79.04 K and test MAE is 52.02 K, but test $n = 6$ makes the test number unstable. More important, simple baselines are competitive or superior.

Table 4: Control baselines compared with the Tg Transformer. MAE is in Kelvin.

Model	Validation MAE	Test MAE
Mean predictor	88.48	132.54
Median predictor	87.53	130.67
Length-only linear regression	76.77	76.39
Bag-of-tokens ridge	79.59	43.24
Transformer	79.04	52.02

The uncomfortable interpretation is that the Transformer is not clearly learning chemistry beyond shallow representation signals. A length-only model beating it on validation is not a minor nuisance; it changes how every adversarial-drift number should be interpreted.

4.2 Forensic Audit Findings

4.3 Padding and Masking Are Correct

The audit tested whether padding artifacts caused the drift. Appending masked PAD positions produced exactly zero drift:

Table 5: Masked padding no-op test.

Extra masked pads	0	5	20	100
Max absolute drift (K)	0.0000	0.0000	0.0000	0.0000

This refutes a simple masking or batching bug. The instability is not caused by masked padding being included in the pooled representation.

4.4 Position Shift Alone Is Too Small

Appending unmasked PAD-like tokens changes length and position indexing without meaningful chemistry. This produced only modest drift.

Table 6: Unmasked pad drift probe.

Unmasked pads appended	1	3	10
Mean absolute drift (K)	0.35	4.99	5.59

Therefore, positional re-indexing is a real effect, but not large enough to explain the full 30 K-scale perturbation drift.

4.5 Length Mechanism Refuted

The decisive control compares length-preserving token reordering with a length-changing duplicate-token edit.

Table 7: Length-preserving controls versus length-changing control across seeds.

Seed	Shuffle	Reverse	Duplicate +1 token
20260815	36.03	38.58	38.66
20260816	40.47	37.13	24.92
20260817	20.07	17.61	18.80
20260818	24.60	23.35	18.68
20260819	39.07	35.51	23.72
Mean	32.05	30.44	24.96

Shuffle and reverse preserve length and token multiset, yet they produce drift matching or exceeding the length-changing duplicate edit. Thus, the old claim that length-changing perturbations cause the vulnerability is false. The model is unstable to token-level restructuring in general.

4.6 Mean-Pool Dilution Confirmed

Single-token edit drift decreases with sequence length, as expected from masked mean pooling.

Table 8: Drift by token-length bin.

Token length bin	n	Insertion drift (K)	Deletion drift (K)
[0, 15)	3	75.34	46.12
[15, 30)	8	54.72	40.11
[30, 60)	18	26.76	17.41
[60, 999)	8	19.25	22.01

The correlation between length and insertion drift is -0.502 , while the correlation between inverse length and insertion drift is $+0.575$. This does not mean pooling is a software bug. It means the selected representation compresses sequences in a way that makes drift depend on L .

4.7 Attack Families Are Not Comparable

The chemical-severity audit shows that the attack families do not represent matched interventions.

Table 9: Phase 2B valid candidate chemistry audit.

Family	Valid n	Validity	Mean $ \Delta MW $	Formula kept	Identical	Mean drift
Substitution	88	48.6%	10.79	0.0%	0.0%	12.97 K
Insertion	85	45.9%	15.75	12.9%	12.9%	30.67 K
Deletion	94	50.8%	12.28	2.1%	1.1%	30.73 K
Rearrangement	63	63.0%	0.00	100.0%	1.6%	7.99 K

Insertion and deletion often change heavy atoms and molecular formula. Rearrangement preserves formula but may change connectivity. None of these is a clean representation-level attack. Therefore, family comparisons confound mechanism, chemical severity, and validity filtering.

4.8 Defense Results

4.9 Bandgap Phase 2 Defense

The older bandgap defense result is positive but local. Adversarial training reduced seen-family attack success while preserving clean test MAE.

Table 10: Older bandgap Phase 2 comparative evaluation.

Metric	Baseline	Defended	Interpretation
Clean MAE	0.4619 eV	0.4601 eV	Preserved clean accuracy
Substitution success	20.13%	7.53%	Improved on seen family
Rearrangement success	9.29%	2.83%	Improved on seen family
Insertion success	32.61%	32.46%	No transfer
Deletion success	32.13%	30.04%	Weak/no transfer
Mean adversarial drift	0.4655 eV	0.4290 eV	Slight aggregate reduction
Max adversarial drift	5.5894 eV	5.5167 eV	Outliers remain

This is best described as attack-specific robustness, not general robust materials representation learning.

4.10 Tg Defense Interpretation

For the Tg audit, adversarial training may reduce local drift on candidate sets similar to those used during training, but the chemical label-preservation assumption remains unresolved. If an adversarial example is actually a different polymer with a different true Tg, training on the original label is not a clean robustness intervention.

5 Discussion

5.1 Target-Preservation Taxonomy

Table 11: Revised attack taxonomy for Tg robustness claims.

Class	Families	Interpretation
Representation-level	None currently proven in the main attack families	Same molecule, same target label; this is the desired robustness control
Chemistry-changing	Substitution, insertion, deletion	Formula or heavy-atom content can change; true Tg may change
Isomerization / connectivity-changing	Rearrangement	Formula may be preserved, but molecular graph can change
Invalid	Filtered candidates	Not evaluated as valid adversarial examples

The missing representation-level control is SMILES randomization:

same molecular graph $\rightarrow$ different valid string.

If the prediction changes under this transformation, the error is clearly a representation-invariance failure because the target label should be unchanged.

5.2 Bio-Cyber Benchmark as Teaching Scaffold

The synthetic Bio-Cyber benchmark is useful because it teaches the basic adversarial-learning loop in a simpler setting:

sequence $\rightarrow$ encoder $\rightarrow$ model $\rightarrow$ prediction $\rightarrow$ perturbation $\rightarrow$ robustness question.

Its 1D CNN uses one-hot encoding, convolutional filters, max pooling, dropout, and dense layers. Because the implanted motifs are synthetic and controlled, the benchmark is safe and interpretable. However, it should remain clearly separated from the materials project:

- Bio-Cyber is synthetic classification.
- Materials Adversarial is polymer-property regression.
- Bio-Cyber motifs are artificial labels.
- Materials targets are physical properties such as Tg or bandgap.
- Bio-Cyber explains the adversarial-learning idea; Materials is the scientific robustness case study.

5.3 Revised Scientific Claims

5.4 Claims That Survive

1. The project implements a real adversarial evaluation framework for sequence-based polymer property models.

2. The Tg Transformer is unstable under token-level perturbations.
3. The instability is not specifically caused by length-changing edits.
4. Masking, padding, and batching are not the source of the observed drift.
5. Mean pooling creates a measurable length-dependent drift pattern.
6. Attack-family comparisons must account for chemical severity, validity rate, and candidate budget.
7. Bandgap adversarial training demonstrated local improvement against seen attack families, but not broad generalization.

5.5 Claims That Should Not Be Made Yet

1. The Transformer has learned robust polymer chemistry.
2. Length-changing perturbations are the causal mechanism of vulnerability.
3. Current token edits preserve Tg.
4. A six-sample test set supports strong clean-MAE conclusions.
5. Adversarial training proves chemistry-preserving robustness.
6. Maximum drift comparisons are reliable without matched candidate budgets.

6 Limitations and Future Work

6.1 Priority 1: SMILES Randomization Invariance

Generate multiple randomized SMILES for each molecule using RDKit while preserving the same molecular graph. Evaluate:

$$\Delta_{\text{rand}}(x) = \max_i |f(r_i(x)) - f(x)|.$$

Report mean, median, maximum, and per-sample distributions. This experiment directly measures representation invariance without changing the target label.

6.2 Priority 2: Positional-Encoding Ablation

Retrain the Transformer with no positional embedding, or with alternative position schemes, across five seeds. Compare clean MAE and drift under:

- insertion;
- deletion;
- shuffle/reverse controls;
- SMILES randomization.

This isolates whether learned absolute positions are a small contributor or a central mechanism.

6.3 Priority 3: Cross-Validated Baseline Gate

Run scaffold cross-validation for:

- mean predictor;
- median predictor;
- length-only regression;
- bag-of-tokens ridge;
- Transformer;
- optionally descriptor-based classical baselines.

The Transformer should not be treated as scientifically meaningful unless it beats shallow controls with error bars.

6.4 Priority 4: Drift Residualization

Model drift as a function of inverse length:

$$\Delta = \beta_0 + \beta_1 \frac{1}{L} + \epsilon.$$

Then report residual drift. This separates model instability from the predictable compression artifact induced by mean pooling.

6.5 Priority 5: Budget-Matched Attack Evaluation

For each attack family, evaluate the same number of valid candidates per sample or use a matched sampling protocol. Report:

- candidate budget;
- validity rate;
- no-op rate after canonicalization;
- median drift;
- mean drift;
- maximum drift with confidence intervals or bootstrapping.

6.6 Recommended Next Experiments

The strongest honest abstract would say:

We developed an adversarial evaluation and defense framework for polymer sequence property models. Initial attacks revealed large prediction drift, but forensic controls showed that the apparent length-change mechanism was incorrect. Length-preserving token reordering caused comparable drift, simple length and bag-of-token baselines matched or exceeded the Transformer on the small Tg split, and current token-edit attacks did not establish label-preserving chemical robustness. These findings reposition the contribution from “a solved defense” to a rigorous audit of failure modes in sequence-based materials learning, motivating representation-level controls such as SMILES randomization and cross-validated shallow baselines.

6.7 Publication-Ready Framing

The project is valuable precisely because it did not merely produce a neat defense number. It built a pipeline, generated adversarial evidence, and then audited its own conclusions hard enough to overturn the attractive but false length-mechanism story. The current scientific position is:

instability confirmed + length mechanism refuted + label-preserving robustness still unproven.

That is a publishable kind of honesty if the next experiments are framed as controls rather than as cosmetic additions.

7 Conclusion

The forensic audit supports a narrower and more defensible conclusion than the original length-change narrative. Polymer sequence models can exhibit large prediction changes under valid-looking token perturbations, but length change is not sufficient to explain that behavior: length-preserving reorderings are also highly disruptive. The observed vulnerability is therefore best described as token-level representation instability, with contributions from learned absolute positions, masked mean pooling, small-data estimation, and dataset shift.

Adversarial training reduces drift for some attack families, especially when candidate sets are controlled, but the result should not be interpreted as proof that the model has learned chemical invariance. The present attacks do not consistently preserve molecular identity or the target property. A strong robustness claim requires an exact-graph representation control, such as canonicalization-verified SMILES randomization, together with matched budgets, cross-validated baselines, and uncertainty-aware reporting.

A Quick Reference Tables

Table 12: High-level contrast between the repository’s two sequence projects.

Dimension	Bio-Cyber benchmark	Materials adversarial project
Data type	Synthetic DNA-like strings	Polymer PSMILES strings
Task	Classification	Regression
Target	Class 0 / Class 1 synthetic motif label	Tg in K or bandgap in eV, depending on milestone
Model	1D CNN	Transformer regressor
Main use	Safe teaching and baseline adversarial setup	Scientific robustness audit
Key caution	Not a real biological system	Valid string is not proof of same material property

Table 13: One-page verdict matrix for Tg forensic audit.

Issue	Evidence	Verdict
Padding/masking bug	Masked pads produce zero drift	Refuted
Position shift	Unmasked +10 pads produce 5.59 K drift	Real but small
Length mechanism	Shuffle/reverse match duplicate-token drift	Refuted
Mean-pool dilution	Drift correlates with $1/L$	Confirmed
Chemistry-preserving attacks	Insertion/deletion change formula; rearrangement changes connectivity	Not established
Model chemistry learning	Length-only model beats Transformer on validation	Not demonstrated
Defense generalization	Seen-family gains do not transfer cleanly	Limited

Data and Code Availability

The analyses summarized here are grounded in the local project repository. The source files, result manifests, diagnostic tables, and generated figures should be archived with a version identifier before submission so that the reported counts and split assignments can be independently reproduced.

References

References

- [1] Greg Landrum et al. RDKit: Open-source cheminformatics. Available at <https://www.rdkit.org/>.
- [2] Ashish Vaswani et al. Attention is all you need. In *Advances in Neural Information Processing Systems*, 2017.
- [3] Weihua Jin, Regina Barzilay, and Tommi Jaakkola. Junction tree variational autoencoder for molecular graph generation. In *Proceedings of ICML*, 2018.
- [4] Philippe Schwaller et al. Molecular transformer: A model for uncertainty- calibrated chemical reaction prediction. *ACS Central Science*, 5(9), 2019.
- [5] Materials Adversarial Learning Project. *Forensic diagnostics, audit reports, and architecture notes*. Local repository artifacts, 2026.